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Aryabhatta Knowledge University, Patna

B.Tech 3rd Semester Exam., 2019

Time: 03 Hours

Code: 101308 ***Subject: Surveying and Geomatics*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer (any seven): **[2x7=14]**

(a) On a turning point in levelling operation
(i) only a backsight is taken
(ii) only a foresight is taken
(iii) Both (i) and (ii)
(iv) an intersight is taken

(b) Method of repetition for angle measurement can be used in measurement of
(i) vertical angle
(ii) horizontal angle
(iii) Both (i) and (ii)
(iv) None of the above

(c) Cross-staff is used to set
(i) acute angle
(ii) obtuse angle
(iii) staff reading
(iv) right angle

(d) A negative declination shows that the magnetic meridian is to the
(i) West of true meridian
(ii) East of true meridian
(iii) South of true meridian
(iv) North of true meridian

(e) The curvature error and atmospheric refractions are eliminated by
(i) simple levelling
(ii) differential levelling
(iii) stadia levelling
(iv) reciprocal levelling

(f) To compute area of a figure bounded by straight lines, the figure is generally converted into a network of
(i) squares
(ii) triangles
(iii) rectangles
(iv) trapezium

(g) Which one of the following factors does not affect the relief displacement of an object in aerial photographs?
(i) Focal length
(ii) Flying height
(iii) Height of the object
(iv) Type of aerial camera

(h) The altitudinal distance of the GPS satellites from the earth is about
(i) 20200 km
(ii) 30000 km
(iii) 36000 km
(iv) 44000 km

(i) Which of the following statements is not correct concerning refraction of the GPS signal in the troposphere?
(i) L1 and L2 carrier waves are refracted equally.
(ii) When a GPS satellite is near the horizon, its signal is most affected by the atmosphere.
(iii) The density of the troposphere governs the severity of its effect on a GPS signal.
(iv) The wet component of refraction in the troposphere contributes the larger portion of the range error.

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Q.2 Solve both questions : **[14]**

(a) Discuss different types of levelling with the help of suitable diagram(s).

(b) A dumpy level is set near to station A and the staff readings at A and at the distant station B are 0.95 m and 2.87 m respectively. When the instrument is set near to station B, the staff readings are 0.46 m at A and 1.01 m at B. If AB is 300 m, then determine the collimation error in degrees. Also by assuming a reasonable value of radius of earth, estimate the curvature error.

Q.3 Solve both questions : **[14]**

(a) What are different types of bearing? Discuss the method of converting a whole circle bearing to reduced bearing.

(b) Discuss the different instruments used in plane table surveying. Also explain the method of intersection in plane table surveying.

Q.4 Solve all questions : **[14]**

(a) Explain the temporary adjustments for angle measurement using Vernier theodolite.

(b) What do you mean by baseline in triangulation? Explain in detail method to set out the baseline.

(c) Two stations A and B at 90 km apart have elevation of 190 m and 980 m respectively. If the elevation of a peak P at a distance 37 km from A is 280 m, then using Captain McCaw's method find out whether stations A and B are inter-visible or not.

Q.5 Solve both questions : **[14]**

(a) Draw a neat diagram of a simple circular curve. With the help of the diagram, explain the various elements of simple circular curve.

(b) Two tangents intersect at the chainage of 2000 m, the deflection angle being 36°. Calculate all the data necessary for setting out a circular curve with radius of 300 m by deflection angle method. The peg interval is 30 m.

Q.6 Solve both questions : **[14]**

(a) Differentiate between Geodimeter and Tellurometer. What are different factors that affect the accuracy of distance measurement using EDM instruments?

(b) Discuss the basic components of a total station. Discuss in detail about the different applications of total station for civil engineering projects.

Q.7 Solve both questions : **[14]**

(a) What is vertical aerial photograph? With respect to a vertical aerial photograph, discuss the following using a suitable diagram: (i) Principal Point (ii) Principal Line (iii) Isocentre (iv) Camera Axis (v) Nadir Point

(b) A tower was identified on a vertical photograph and the radial distance between its top and its bottom was measured to be 15 mm. The photo-coordinates of top of the displaced tower are x_a= 45mm and y_a= 72mm. The focal length used to take this photograph is 305 mm. If the flying height of the aircraft is 1500 m above the mean sea level (MSL) and the base of the building is 400 m above MSL, how tall is the building?

Q.8 Solve both questions : **[14]**

(a) What do you understand by GPS? How is it helpful in mapping of a region?

(b) What are the different methods for GPS surveying? Explain any one of them in detail.

Q.9 Solve both questions : **[14]**

(a) Describe various applications of remote sensing in the field of civil engineering with suitable examples.

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(b) Discuss in brief the digital image processing process. What are the advantages of digital image processing?

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